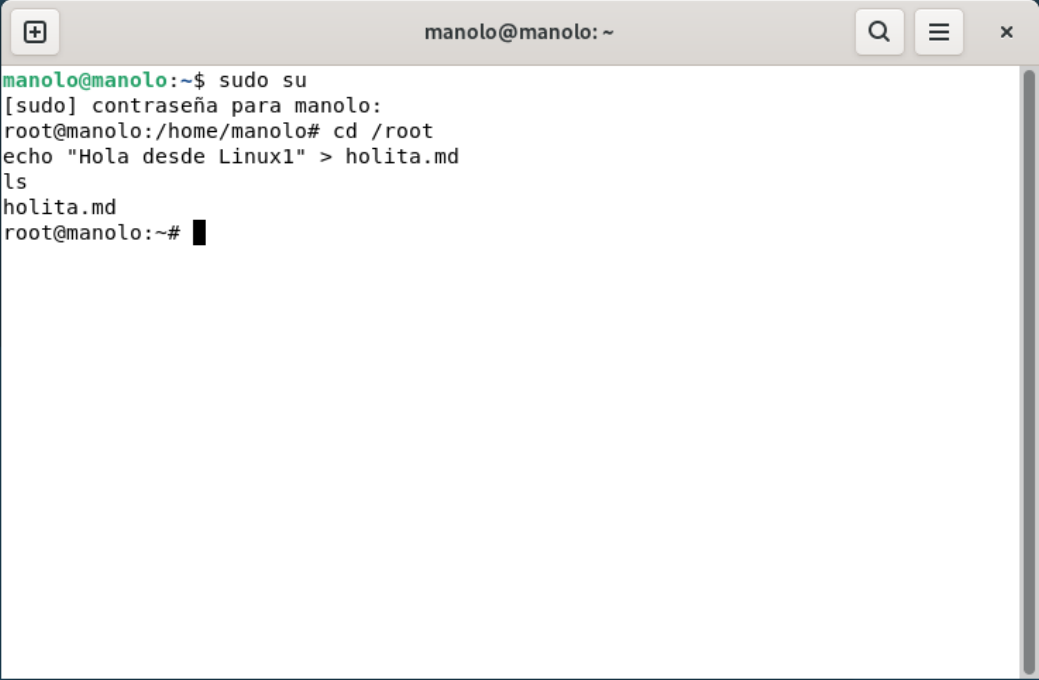


PRACTICA 2: Introduccion al scripting en bash y transferencia de medios por script

Instalacion:

Primero de todo hacemos un sudo su para entrar como root y creamos el primer script



```
manolo@manolo:~$ sudo su
[sudo] contraseña para manolo:
root@manolo:/home/manolo# cd /root
root@manolo:/root# echo "Hola desde Linux1" > holita.md
root@manolo:/root# ls
holita.md
root@manolo:/root#
```

Ahora en la maquina 1 instalamos los paquetes ssh

```
manolo@manolo: ~  
Preparando para desempaquetar .../libcap2-bin_1%3a2.75-10+b3_amd64.deb ...  
Desempaquetando libcap2-bin (1:2.75-10+b3) sobre (1:2.75-10+b1) ...  
Preparando para desempaquetar .../libss2_1.47.2-3+b7_amd64.deb ...  
Desempaquetando libss2:amd64 (1.47.2-3+b7) sobre (1.47.2-3+b3) ...  
Preparando para desempaquetar .../openssl_3.5.4-1~deb13u2_amd64.deb ...  
Desempaquetando openssl (3.5.4-1~deb13u2) sobre (3.5.4-1~deb13u1) ...  
Configurando libcap2-bin (1:2.75-10+b3) ...  
Configurando libss2:amd64 (1.47.2-3+b7) ...  
Configurando openssl (3.5.4-1~deb13u2) ...  
Procesando disparadores para man-db (2.13.1-1) ...  
Procesando disparadores para libc-bin (2.41-12+deb13u1) ...  
Scanning processes...  
Scanning candidates...  
Scanning linux images...  
  
Restarting services...  
systemctl restart apache2.service  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
root@manolo:/home/manolo#
```

En la maquina 2 me saltaron varios errores, pero se solucionaron

```
root@manolo-virtual-machine: /home/manolo  
3072 SHA256:Fob5IgZjcustB3FvSFH8Bgq6K7tLRnMQVhVoxrJK/pA root@manolo-virtual-mach  
ine (RSA)  
Creating SSH2 ECDSA key; this may take some time ...  
256 SHA256:Xv6oQclbhr8qEyPlu66JDQLCPK7luN4Kb534KuvHKEM root@manolo-virtual-machi  
ne (ECDSA)  
Creating SSH2 ED25519 key; this may take some time ...  
256 SHA256:Ilrdfufj9yGsBGr5p7jHD6dwxYCSO/nlvLAAaRUd2XY root@manolo-virtual-machi  
ne (ED25519)  
Created symlink /etc/systemd/system/ssh.service → /lib/systemd/system/ssh.servi  
ce.  
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /lib/s  
ystemd/system/ssh.service.  
rescue-ssh.target is a disabled or a static unit, not starting it.  
ssh.socket is a disabled or a static unit, not starting it.  
Setting up ssh-import-id (5.11-0ubuntu1) ...  
Setting up ncurses-term (6.3-2ubuntu0.1) ...  
Processing triggers for man-db (2.10.2-1) ...  
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...  
root@manolo-virtual-machine:/home/manolo# sudo systemctl enable ssh  
sudo systemctl start ssh  
Synchronizing state of ssh.service with SysV service script with /lib/systemd/sy  
stemd-sysv-install.  
Executing: /lib/systemd/systemd-sysv-install enable ssh  
root@manolo-virtual-machine:/home/manolo#
```

Y aqui se ve como estan ambas maquinas ssh activas

```
manolo@manolo: ~  
root@manolo:/home/manolo# sudo systemctl enable ssh  
sudo systemctl start ssh  
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/sy  
stemd-sysv-install.  
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh  
root@manolo:/home/manolo# systemctl status ssh  
● ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enab>  
   Active: active (running) since Tue 2026-02-03 12:41:31 CET; 16min ago  
  Invocation: 0372d919bae2421a9923429968c12dc8  
     Docs: man:sshd(8)  
           man:sshd_config(5)  
  Main PID: 1354 (sshd)  
    Tasks: 1 (limit: 7076)  
  Memory: 2.6M (peak: 3.2M)  
     CPU: 158ms  
   CGroup: /system.slice/ssh.service  
           └─1354 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
feb 03 12:41:30 manolo systemd[1]: Starting ssh.service - OpenBSD Secure Shell  
feb 03 12:41:31 manolo sshd[1354]: Server listening on 0.0.0.0 port 22.  
feb 03 12:41:31 manolo systemd[1]: Started ssh.service - OpenBSD Secure Shell s  
feb 03 12:41:31 manolo sshd[1354]: Server listening on :: port 22.  
lines 1-17/17 (END)
```

```
root@manolo:/home/manolo# sudo systemctl start ssh  
Synchronizing state of ssh.service with SysV service script with /lib/systemd/sy  
stemd-sysv-install.  
Executing: /lib/systemd/systemd-sysv-install enable ssh  
root@manolo-virtual-machine:/home/manolo# systemctl status ssh  
● ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: e>  
   Active: active (running) since Tue 2026-02-03 12:52:31 CET; 6min ago  
     Docs: man:sshd(8)  
           man:sshd_config(5)  
  Main PID: 3071 (sshd)  
    Tasks: 1 (limit: 9373)  
  Memory: 1.7M  
     CPU: 59ms  
   CGroup: /system.slice/ssh.service  
           └─3071 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
feb 03 12:52:31 manolo-virtual-machine systemd[1]: Starting OpenBSD Secure Shel  
feb 03 12:52:31 manolo-virtual-machine sshd[3071]: Server listening on 0.0.0.0  
feb 03 12:52:31 manolo-virtual-machine sshd[3071]: Server listening on :: port  
feb 03 12:52:31 manolo-virtual-machine systemd[1]: Started OpenBSD Secure Shel  
lines 1-16/16 (END)
```

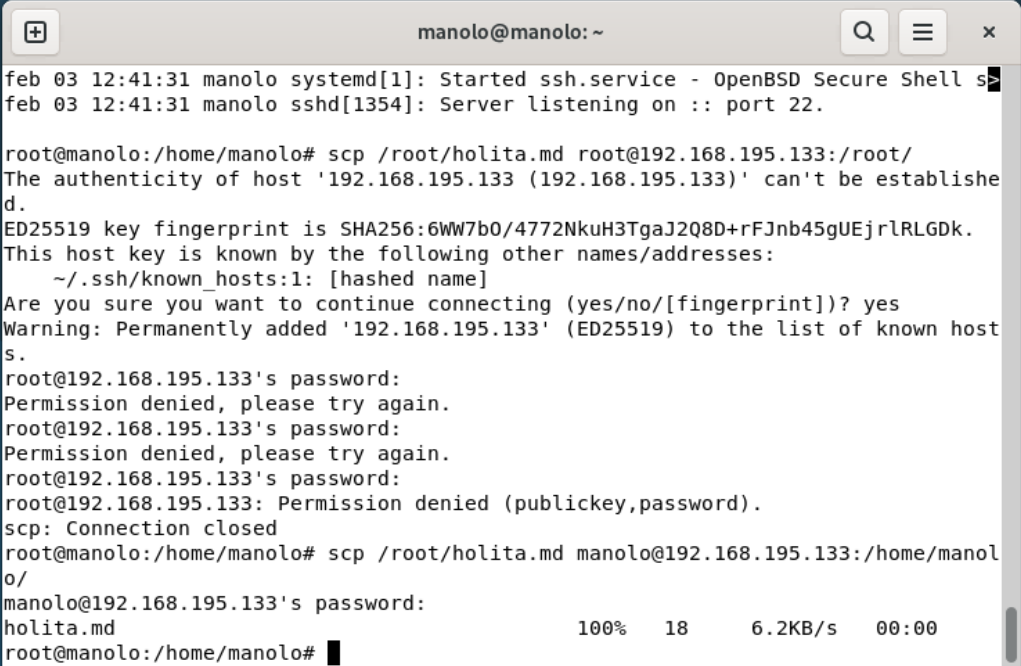
Ahora hago ping y la máquina recibe los paquetes

```
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.860/1.008/1.157/0.148 ms
root@manolo-virtual-machine:/home/manolo# ping 192.168.195.133
PING 192.168.195.133 (192.168.195.133) 56(84) bytes of data.
64 bytes from 192.168.195.133: icmp_seq=1 ttl=64 time=0.728 ms
64 bytes from 192.168.195.133: icmp_seq=2 ttl=64 time=0.914 ms
^C
--- 192.168.195.133 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.728/0.821/0.914/0.093 ms
root@manolo-virtual-machine:/home/manolo# ping 192.168.195.132
PING 192.168.195.132 (192.168.195.132) 56(84) bytes of data.
64 bytes from 192.168.195.132: icmp_seq=1 ttl=64 time=0.116 ms
64 bytes from 192.168.195.132: icmp_seq=2 ttl=64 time=0.048 ms
^C
--- 192.168.195.132 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1042ms
rtt min/avg/max/mdev = 0.048/0.082/0.116/0.034 ms
root@manolo-virtual-machine:/home/manolo#
```

Y creamos el script

```
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.728/0.821/0.914/0.093 ms
root@manolo-virtual-machine:/home/manolo# ping 192.168.195.132
PING 192.168.195.132 (192.168.195.132) 56(84) bytes of data.
64 bytes from 192.168.195.132: icmp_seq=1 ttl=64 time=0.116 ms
64 bytes from 192.168.195.132: icmp_seq=2 ttl=64 time=0.048 ms
^C
--- 192.168.195.132 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1042ms
rtt min/avg/max/mdev = 0.048/0.082/0.116/0.034 ms
root@manolo-virtual-machine:/home/manolo# sudo su
cd /root
echo "Hola desde la maquina 1" > holita.md
ls
root@manolo-virtual-machine:/home/manolo#
```

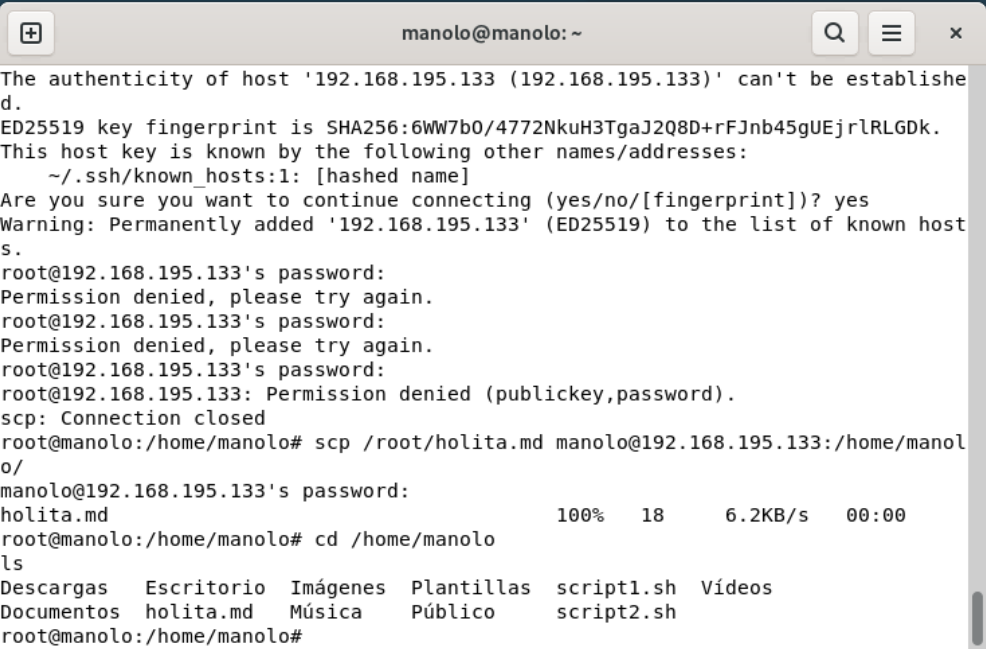

Y vemos como la otra maquina a traves del ssh recibe el script



```
manolo@manolo: ~
feb 03 12:41:31 manolo systemd[1]: Started ssh.service - OpenBSD Secure Shell s
feb 03 12:41:31 manolo sshd[1354]: Server listening on :: port 22.

root@manolo:/home/manolo# scp /root/holita.md root@192.168.195.133:/root/
The authenticity of host '192.168.195.133 (192.168.195.133)' can't be establishe
d.
ED25519 key fingerprint is SHA256:6WW7b0/4772NkuH3TgaJ2Q8D+rFJnb45gUEjrlRLGDK.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.195.133' (ED25519) to the list of known host
s.
root@192.168.195.133's password:
Permission denied, please try again.
root@192.168.195.133's password:
Permission denied, please try again.
root@192.168.195.133's password:
root@192.168.195.133: Permission denied (publickey,password).
scp: Connection closed
root@manolo:/home/manolo# scp /root/holita.md manolo@192.168.195.133:/home/manol
o/
manolo@192.168.195.133's password:
holita.md                               100%  18    6.2KB/s   00:00
root@manolo:/home/manolo#
```

Y se ve como la tiene guardada



```
manolo@manolo: ~
The authenticity of host '192.168.195.133 (192.168.195.133)' can't be establishe
d.
ED25519 key fingerprint is SHA256:6WW7b0/4772NkuH3TgaJ2Q8D+rFJnb45gUEjrlRLGDK.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.195.133' (ED25519) to the list of known host
s.
root@192.168.195.133's password:
Permission denied, please try again.
root@192.168.195.133's password:
Permission denied, please try again.
root@192.168.195.133's password:
root@192.168.195.133: Permission denied (publickey,password).
scp: Connection closed
root@manolo:/home/manolo# scp /root/holita.md manolo@192.168.195.133:/home/manol
o/
manolo@192.168.195.133's password:
holita.md                               100%  18    6.2KB/s   00:00
root@manolo:/home/manolo# cd /home/manolo
ls
Descargas  Escritorio  Imágenes  Plantillas  script1.sh  Vídeos
Documentos holita.md   Música    Público     script2.sh
root@manolo:/home/manolo#
```

Y activamos el script de la otra maquina a traves de la segunda

```
manolo@manolo: ~  
ED25519 key fingerprint is SHA256:6WW7b0/4772NkuH3TgaJ2Q8D+rFJnb45gUEjrLRLGDk.  
This host key is known by the following other names/addresses:  
  ~/.ssh/known_hosts:1: [hashed name]  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.195.133' (ED25519) to the list of known hosts.  
root@192.168.195.133's password:  
Permission denied, please try again.  
root@192.168.195.133's password:  
Permission denied, please try again.  
root@192.168.195.133's password:  
root@192.168.195.133: Permission denied (publickey,password).  
scp: Connection closed  
root@manolo:/home/manolo# scp /root/holita.md manolo@192.168.195.133:/home/manolo/  
manolo@192.168.195.133's password:  
holita.md                                100% 18      6.2KB/s   00:00  
root@manolo:/home/manolo# cd /home/manolo  
ls  
Descargas  Escritorio  Imágenes  Plantillas  script1.sh  Vídeos  
Documentos holita.md  Música    Público    script2.sh  
root@manolo:/home/manolo# cat holita.md  
Hola desde Linux1  
root@manolo:/home/manolo#
```